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REVIEW

Biology and Biocontrol Potential of *Ampelomyces* Mycoparasites, Natural Antagonists of Powdery Mildew Fungi

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Historically, pycnidial fungi belonging to the genus *Ampelomyces* were among the first mycoparasites to be studied in detail and were also the first fungi used as biocontrol agents of plant parasitic fungi. The interactions between host plants, powdery mildew fungi and *Ampelomyces* mycoparasites are one of the most evident cases of tritrophic relationships in nature although their study has received little attention in fungal and plant ecology so far. *Ampelomyces* mycoparasites have now become one of the most advanced in terms of commercial development of a fungal biocontrol agent, although there is still a need for more development work to produce a product with reliability approaching that of conventional chemical treatments. This review summarizes the taxonomy, genetic diversity, life cycle, mode of action, natural occurrence, host range, biocontrol potential, mass production and commercialization of these mycoparasites and compares the biocontrol ability of *Ampelomyces* with that of other fungal antagonists of powdery mildews.

Keywords: *Ampelomyces* spp., powdery mildew, hyperparasitism, plant protection, integrated control, biocontrol, fungal antagonists, AQ10, tritrophic interactions

INTRODUCTION

The exploitation of antagonistic trophic interactions between plant inhabiting microorganisms offers an opportunity for their use in biological control of plant diseases, yet few examples have reached the market. One notable exception is the use of *Ampelomyces* as a biocontrol agent for powdery mildew. Powdery mildew infections remain among the most important plant pathological problems world-wide, both in agriculture and horticulture,

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despite extensive research on their pathogenesis, epidemiology and control. These diseases account for the greatest proportion of the fungicides applied in European agriculture (Hewitt, 1998). Important crops including wheat (*Triticum aestivum*), barley (*Hordeum vulgare*), grape-vine (*Vitis vinifera*), apple (*Malus domestica*), and several vegetables and ornamentals, grown in the field or greenhouses, are among the major targets of powdery mildew fungi. Cultivars resistant or tolerant to powdery mildew infections have been developed in a number of crops, but their use is limited, especially in fruit and vegetable crops (Bélanger & Benyagoub, 1997). In other significant crops, such as tomato (*Lycopersicon esculentum*), all the cultivars and hybrids available to the growers world-wide are susceptible to powdery mildew (Kiss *et al.*, 2001).

The constraints associated with the use of fungicides and resistant cultivars have led to the search of alternative methods to control powdery mildews. The efficacy of non-fungicide products, such as oils, salts, soluble silicon and plant extracts have extensively been studied, especially in greenhouse production (Bélanger & Benyagoub, 1997; Pasini *et al.*, 1997; McGrath & Shishkoff, 1999). Many potential biocontrol agents of powdery mildews have also been tested. Among these, the use of mycolytic bacteria (Skidmore, 1976; Highland, 2000) and mycophagous arthropods (Yarwood, 1957; English-Loeb *et al.*, 1999) have not so far showed promise of practical control. In contrast, better results have been obtained in biocontrol trials that have used fungi antagonistic to powdery mildews.

Recently, Kiss (2003) reviewed the ecology, modes of action and biocontrol efficacy of the approximately 40 fungal species that have been reported as natural antagonists of powdery mildews or have been tested as potential biocontrol agents. Of these, pycnidial fungi belonging to the genus *Ampelomyces* Ces. are the oldest known and the commonest natural antagonists of powdery mildews that have been intensively studied in crop protection practice. One *Ampelomyces* strain has already been formulated, registered and commercialized in some countries by Ecogen, Inc. (Langhorne, PA, USA) as a product under the trade name AQ10™ Biofungicide. A second biofungicide, Sporodex®, a product containing a strain of *Pseudozyma flocculosa* (Traquair, Shaw & Jarvis) Boekhout & Traquair, is now also available for use against powdery mildews (Paulitz & Bélanger, 2001; Bélanger & Labbé, 2002). AQ10 Biofungicide is one of the few biocontrol products developed against fungal plant pathogens (Whipps & Lumsden, 2001), with the result that *Ampelomyces* mycoparasites are among the most studied biocontrol agents. In addition, the interactions between host plants, powdery mildew fungi and *Ampelomyces* mycoparasites are one of the most evident cases of tritrophic relationships in nature, because this relationship is common world-wide and takes place exclusively on aerial plant surfaces, thus facilitating its direct observation (Kiss, 1998). However, it has received little attention in fungal and plant ecology, although it could be used as a model to study the significance of mycoparasitism in the natural dynamics of plant parasitic fungi.

This review is concentrated on the biology and biocontrol potential of *Ampelomyces* spp. to provide an insight for future research in fungal and plant ecology as well as in the development of products for control of plant diseases. It is timely as there is controversy regarding the identification of pycnidial mycoparasites of powdery mildews (Sullivan & White, 2000). It should be noted that major changes have recently occurred in the taxonomy and nomenclature of the Erysiphaceae (Braun *et al.*, 2002). However, the names of powdery mildew fungi used in this review are identical with those used in the corresponding references to avoid confusion. Plant names also follow the matching references.

REVIEW OF LITERATURE

Early Studies of *Ampelomyces* Mycoparasites

In 1852, a pycnidial fungus associated with powdery mildew colonies covering grape-vine leaves was described succinctly as *Ampelomyces quisqualis* Ces. (Cesati, 1852). Since then, other similar fungi have been reported by a number of mycologists under various names,

such as *Byssocystis textilis* Riess (Riess, 1852), *Cicinobolus florentinus* Ehrenb. (Ehrenberg, 1853), *Cicinnobolus cesatii* (De Bary, 1870), *Cicinobolus humuli* Faut. (Fautrey, 1890). Some authors (e.g., Cesati, 1852; Riess, 1852; De Bary, 1870) clearly recognized that these fungi were parasites of powdery mildews (Figure 1a,b), but others (e.g., von Mohl, 1852; Tulasne, 1856) considered that their pycnidia and conidia were part of the life cycle of the powdery mildews with which they were found (Figure 2). The first thorough study of these fungi was carried out by De Bary (1870) who showed that the hyphae of *Ampelomyces* grow internally in the mycelia of powdery mildews from cell to cell, and produce their pycnidia in one or two cells of the hyphae, conidiophores (Figure 1b), conidium initials and ascocarps of their fungal hosts. De Bary (1870) proved experimentally that the intracellular hyphae can grow out from the parasitized cells when placed in water for a few hours (Figure 1a). In addition, De Bary (1870) carried out cross-inoculation experiments demonstrating that an *Ampelomyces* mycoparasite collected from a given powdery mildew species could also produce intracellular pycnidia in mycelia of other powdery mildew species. His pioneering work was the first detailed study published of an interfungal parasitic relationship. Emmons (1930) subsequently published an extensive cytological study describing in detail the penetration, growth and sporulation of *Ampelomyces* in the ascocarps of powdery mildews. The potential for biocontrol was realized around the same time and Yarwood (1932) reported the treatment of powdery mildew infected plants with a conidial suspension of *Ampelomyces*, thus carrying out the first biocontrol experiment against a plant pathogenic fungus. Other early literature was mainly observational (e.g., Fautrey, 1890; Griffiths, 1899; Hino and Kato, 1929; Clare, 1964; Da Ponte, 1966), and reported only the occurrence of *Ampelomyces* in various powdery mildews rather than adding to the knowledge regarding biology, physiology or biocontrol potential.

Taxonomy, Nomenclature and Genetic Diversity of Pycnidial Mycoparasites of Powdery Mildews

Taxonomic controversies appeared soon after the genus *Ampelomyces* was erected by Cesati (1852). The generic name *Cicinnobolus* was commonly used until the 1970s (Donk, 1966) despite the priority of the name *Ampelomyces* being noted by Rogers (1959) and Speer (1978a). In addition, some authors split the genus in more than 40 taxa describing new species, formae and new combinations (comb. nov.) based mainly on the host plants or host powdery mildews in which the mycoparasites were found (e.g., Da Ponte, 1966; Rudakov,

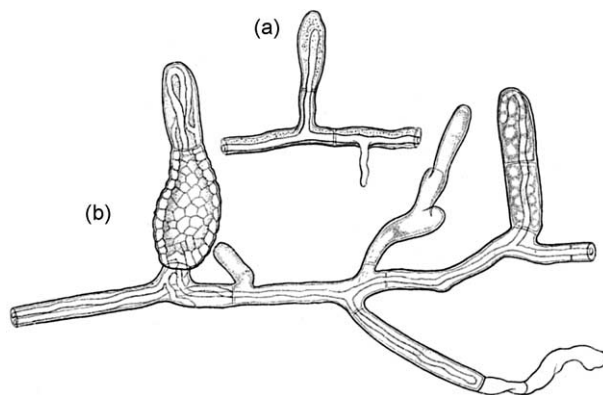


FIGURE 1. De Bary's drawings from 1870. (a) An intracellular hypha of *Ampelomyces* growing out from the powdery mildew mycelium. (b) An intracellular pycnidium of *Ampelomyces* in a powdery mildew conidiophore.

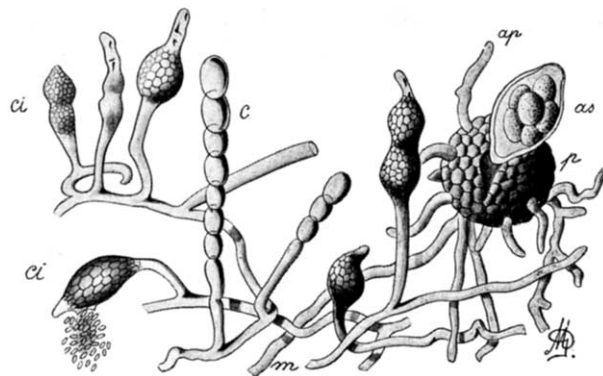


FIGURE 2. Tulasne's drawings redrawn by Moesz (1912) showing the pycnidia of *Ampelomyces* as one of the fruiting bodies of powdery mildews.

1979). The descriptions of these species and other taxa are formally still valid, although their names are not currently in use as there is no evidence for host specificity in *Ampelomyces* (De Bary, 1870; Blumer, 1933, 1967; Philipp & Crüger, 1979; Kranz, 1981; Sztejnberg *et al.*, 1989; Kiss & Vajna, 1995). Consequently, current practice is to consider that the genus is monospecific, and that *A. quisqualis* is a common pycnidial intracellular mycoparasite of all powdery mildews world-wide. However, in the absence of a formal taxonomic reassessment of the many validly described species and other taxa, the use of the binomial '*A. quisqualis*' for all the pycnidial hyperparasites of powdery mildew fungi is not technically correct and a taxonomic revision is clearly needed (Sutton, 1980; Kranz, 1981).

This situation is further complicated by a variation in the morphology and cultural patterns of different '*Ampelomyces*' isolates, and this has suggested that the binomial '*A. quisqualis*' has been applied to a range of isolates representing a species complex (Kiss & Vajna, 1995). However, in the absence of detailed studies, these fungi will still be described under the '*Ampelomyces*' epithet (Kiss, 1997a). Differences were reported in isolates of pycnidial mycoparasites of powdery mildews in the morphology of conidia (Speer, 1978b; Rudakov, 1979), pycnidia (Clare, 1964; Belsare *et al.*, 1980; Shin, 1994), cultural characteristics (Yarwood, 1932; Mhaskar, 1974; Mhaskar & Rao, 1974; Rudakov, 1979; Sz. Nagy & Vajna, 1990; Kiss, 1997a), and rDNA ITS sequences (Kiss, 1997a; Kiss & Nakasone, 1998; Sullivan & White, 2000). Based on the growth rate of these isolates in culture, two types of growth can be distinguished easily under various cultural conditions. Firstly, there are fast growing isolates which extend at 3–4 mm radial growth/day⁻¹ at room temperature. Secondly, there are slower-growing isolates with a radial growth rate of 0.1–1 mm day⁻¹ (Mhaskar, 1974; Rudakov, 1979; Sz. Nagy & Vajna, 1990; Kiss & Vajna, 1995). An RFLP analysis showed high genetic diversity in the rDNA ITS region of 46 *Ampelomyces* isolates obtained from different parts of the world (Kiss, 1997a). Fast growing and slow growing isolates were distinguished by different RFLP patterns. A molecular phylogenetic study of rDNA ITS sequences obtained from ten isolates supported the RFLP analysis, and revealed that slow growing and fast growing isolates were separated into two distinct clades which may not even be congeneric (Kiss & Nakasone, 1998). Slow growing isolates were genetically more diverse, while fast growing ones showed a close phylogenetic relationship with *Epicoccum nigrum* (Kiss & Nakasone, 1998) which is known to have a pycnidial, *Phoma*-like state (Arenal *et al.*, 2000).

Recently, Sullivan & White (2000) suggested the possibility that at least some of the fast growing '*Ampelomyces*' isolates were confused with *Phoma glomerata* (Cda) Wollenw. isolates. These authors found that fast growing isolates typically came from sessile pycnidia

found on mildew-infected leaves, whilst the slow growing isolates came from intracellular pycnidia. ITS1 sequence analysis demonstrated that the fast growing isolates clustered in a clade typified by *Phoma* species, whilst the slow growing isolates were distinct (Sullivan & White, 2000). This study supported the results obtained by Kiss and Nakasone (1998). It seems likely that the fast growing isolates are, in fact, *Phoma* species, whilst the true *Ampelomyces* isolates are always slow growing in culture and always produce intracellular pycnidia in powdery mildew mycelia. A more thorough taxonomic investigation is warranted, as some previous authors have identified fast growing isolates as *Ampelomyces* spp. (e.g., Linnemann, 1968; Mhaskar, 1974; Rudakov, 1979; Sz. Nagy & Vajna, 1990; Kiss, 1997a). *Ampelomyces quercinus*, *A. humuli* and *A. heraclei*, for example, are species names based on fast growing isolates (Rudakov, 1979), but these fungi fall within the *Phoma*/*Epicoccum* clade on the basis of phylogenetic analyses (Kiss & Nakasone, 1998; Sullivan & White, 2000).

Life Cycle

Conidia of true *Ampelomyces* isolates are produced in pycnidia developed intracellularly in the mycelia of powdery mildew fungi (Figures 1b and 2). Conidia are unicellular, hyaline, mostly guttulate and are embedded in a mucilaginous matrix inside the pycnidia. In the presence of water, these matrices swell and conidia are released from intracellular pycnidia as a cirrus by the rupture of the pycnidial wall. In *ca.* 10–20 h under conditions of high humidity, conidia germinate and the hyphae of the mycoparasites can then penetrate the hyphae of powdery mildews in their vicinity. The concentration of *Ampelomyces* conidia on the leaves is also relevant as Gu and Ko (1997) showed that germination rapidly decreases above a concentration of 10^6 cfu ml⁻¹ due to the production of a self inhibitor. After penetration, the hyphae of the mycoparasite continue their growth internally and produce their intracellular pycnidia after 5–8 days in the mycelia of their host fungi (Figure 1b). A high RH of the environment enhances the internal growth and sporulation of *Ampelomyces* (Jarvis & Slingsby, 1977; Philipp & Crüger, 1979).

Conidia can be dispersed within the plant canopy by rain-splash or water run off from plant surfaces. *Ampelomyces* can also spread to long distances as hyphal fragments in parasitized and detached powdery mildew conidia (Jarvis & Slingsby, 1977; Speer, 1978b; Sundheim, 1982; Philipp *et al.*, 1984). When these parasitized air-borne conidia land close to any powdery mildew colony under humid conditions, the outgrowing hyphae of *Ampelomyces* can penetrate their mycelia. Thus, *Ampelomyces* can be transported far from its original fungal host and can parasitize other powdery mildew species. After penetration, the hyphae of *Ampelomyces* invade the host mycelia internally, and produce their pycnidia mostly in the conidiophores and young, immature ascocarps of powdery mildews. Occasionally, they also produce pycnidia in the invaded hyphal cells. The life cycle starts again when pycnidia are mature. Emmons (1930) presented equivocal observations regarding a putative teleomorph, but this was not subsequently reported in detail, and there is no other record of a sexual stage in *Ampelomyces*.

Until recently, little was known about the overwintering of *Ampelomyces*. De Bary (1870), Emmons (1930) and Yarwood (1939) found that the fungus can produce pycnidia saprophytically in the senescent or dead plant tissues at the end of the season, and suggested that these structures served as overwintering structures for *Ampelomyces* in the field. However, Falk *et al.* (1995a,b) considered that saprophytic pycnidia produced in leaf debris were not significant for the overwinter survival of *Ampelomyces* in North American vineyards. They showed that pycnidia of *Ampelomyces* survived until the next season mainly in the parasitized ascocarps of *Uncinula necator* produced on the bark of grapevine stocks. Similarly, Marboutie *et al.* (1995) found that pycnidia of *Ampelomyces* overwintered in the parasitized ascocarps of *Sphaerotheca pannosa* on the bark of peach trees. Conidia of *Ampelomyces* are probably rain dispersed from the parasitized ascocarps in spring, and then

germinate on the young leaves and, if powdery mildew colonies are present, penetrate their hyphae and invade their mycelia (Falk *et al.*, 1995a).

Recently, a detailed study of the overwintering of *Ampelomyces* on apple trees and 13 other woody and herbaceous host plant species revealed that these mycoparasites can survive the winter in many different ways on the host plant surfaces. *In vitro* experiments demonstrated that the overwintered pycnidia of *Ampelomyces* collected from the host plants in the spring, and produced in both the conidiophores and the immature ascomata of powdery mildews during the previous season, can initiate the life cycle of these mycoparasites (Szentiványi & Kiss, 2003). Both conidia found in some of the overwintered pycnidia and cells of the pycnidial walls of empty pycnidia germinated in the spring and gave rise to new intracellular pycnidia of *Ampelomyces* when powdery mildew colonies were inoculated with them *in vitro*. Similar experiments showed that the thick-walled, brownish resting hyphae of *Ampelomyces*, described for the first time by De Bary (1870), can also serve as sources of primary inocula of *Ampelomyces* in the spring (Szentiványi & Kiss, 2003). So, this study demonstrated that the conidia, the pycnidial cells and the cells of the resting hyphae of *Ampelomyces* produced in the mycelia of powdery mildews during the previous season can all initiate the life cycle of these mycoparasites in the spring.

Mode of Action

Some biocontrol fungi have more than one mechanism to antagonize their hosts (Jeffries & Young, 1994). *Trichoderma* spp., for example, produce antifungal compounds, act as mycoparasites, induce plant defence mechanisms and can stimulate plant growth (Howell, 2003). In contrast, *Ampelomyces* acts directly by invasion and destruction of host cytoplasm. *Ampelomyces* kills the parasitized powdery mildew cells by causing a rapid degeneration of the cytoplasm (Hashioka & Nakai, 1980). These mycoparasites suppress both asexual and sexual sporulation of the attacked powdery mildew mycelia by colonizing and destroying the conidiophores, and the immature ascocarps, respectively. The early stage of mycoparasitism is apparently biotrophic, but the invaded cytoplasm then begins to die and a necrotrophic interaction results (Hashioka & Nakai, 1980; Sundheim & Krekling, 1982). Philipp *et al.* (1984) noted that the parasitized powdery mildew colonies can continue their radial growth, but their sporulation stopped soon after *Ampelomyces* penetrated their mycelia. Similarly, Shishkoff and McGrath (2002) showed that *Ampelomyces* could not stop the spread of powdery mildew colonies *in vitro* but did reduce the amount of inoculum produced by each colony. Toxin production has not been detected in *Ampelomyces* (Beuther *et al.*, 1981) in contrast to other pycnidial mycoparasites, such as *Coniothyrium minitans* Campbell (Machida *et al.*, 2001; McQuilken *et al.*, 2002, 2003).

Mycoparasitic attack of powdery mildew mycelia by *Ampelomyces* results in a reduction in the negative effects of the pathogen on the host plant. Abo-Foul *et al.* (1996) showed that infected cucumber plants regained vigour after being treated with *Ampelomyces* which killed the pathogen. Eight days after treatment with a conidial suspension of *Ampelomyces*, chlorophyll content and also CO₂-fixation in the infected cucumber leaves were almost the same as in uninfected controls. These results were recently supported by Romero *et al.* (2003), who found that a treatment with *Ampelomyces* significantly increased the chlorophyll content of detached and mildew-infected melon leaves maintained *in vitro*.

The presence of host fungi is recognized by *Ampelomyces*, and a water-soluble substance from powdery mildew conidia was shown to stimulate the germination of *Ampelomyces* conidia *in vitro* (Gu & Ko, 1997). Directed growth of germ-tubes of *Ampelomyces* towards powdery mildew hyphae has also been observed (Sundheim & Krekling, 1982). As with phytopathogenic fungi, penetration of the host cell wall is likely to involve both enzymatic and mechanical processes. Appressorium-like structures were reported at the point of penetration (Sundheim & Krekling, 1982). Extracellular lytic enzymes have been identified in liquid cultures of *Ampelomyces*, which may play a role in the degradation of the powdery

mildew hyphal walls during penetration (Philipp, 1985). Rotem *et al.* (1999) demonstrated that an *exo*- β -1,3-glucanase is excreted both in culture and during mycoparasitism, and showed that culture filtrates of an *Ampelomyces* isolate could cause degradation of hyphal walls of *Sphaerotheca fusca* in the absence of active mycelium.

Ecological Aspects of the Mode of Action of *Ampelomyces*

In general, when mycoparasites are used as biocontrol agents, a certain level of disease has to be tolerated as they can only attack established infections (Fokkema, 1993). Microbial antagonists which act through antibiosis against powdery mildews, such as *Pseudozyma* spp. or *Tilletiopsis* spp., can kill powdery mildew colonies rapidly, causing complete plasmolysis of their cells (Bélanger & Labbé, 2002). As *Ampelomyces* spp. act against powdery mildews through mycoparasitism, without producing antifungal compounds, they destroy the powdery mildew colonies only slowly, taking 5–7 days. In order to maintain control, the growth and spread of their fungal hosts must be slower than that of *Ampelomyces*. The sporulation rate of powdery mildew colonies depends on the inoculum density, physiological patterns of the host plant, abiotic factors, and so on (Yarwood, 1957; Rouse *et al.*, 1984; Bushnell, 2002). When the sporulation and spread of these pathogens is intense, *Ampelomyces* mycoparasites applied as biocontrol agents cannot control them completely. In these cases, they lag behind the spread of the disease, reducing its severity and limiting its negative effect on the infected plants.

The high RH requirement of *Ampelomyces* (Jarvis & Slingsby, 1977; Philipp & Crüger, 1979; Sundheim, 1982; Falk *et al.*, 1995a,b) represents another limiting factor in its use in biocontrol (Verhaar *et al.*, 1999; Paulitz & Bélanger, 2001). The dry conditions that usually favour the development of powdery mildews are not conducive to the development of *Ampelomyces*.

As powdery mildews are biotrophic pathogens infecting the aerial parts of their host plants, potential biocontrol agents need to be active in the phyllosphere, interacting with the living host plant tissues in a tritrophic system (Kiss, 2001). In general, biocontrol trials of phyllosphere plant pathogenic fungi have been less successful than those against rhizosphere pathogens. The phyllosphere is a much more dynamic environment as the leaves have a limited lifetime compared to roots (Fokkema & Schippers, 1986; Andrews 1990, 1992). Thus, mycoparasites have a shorter window of attack.

Under natural circumstances, *A. quisqualis* may not be observed until late in the growing season on many crops, long after powdery mildews have infected the aerial parts of their host plants (Gadoury & Pearson, 1988; Rankovic, 1997; Kiss, 1998). Thus, powdery mildew epidemics usually reach damaging levels before their growth and sporulation are arrested by *A. quisqualis* (Gadoury & Pearson, 1988; Falk *et al.*, 1995a,b). Since the early dispersal and establishment of the mycoparasite onto the first formed mildew colonies is the most important factor in the degree of suppression of powdery mildew on leaves and fruit (Falk *et al.*, 1995b), application must be targeted to ensure earlier infection of the powdery mildew pathogens. However, it should be remembered that, by definition, biotrophic pathogens sporulate on living host tissue. This implies that some level of disease must be tolerated when mycoparasites are used for control (Fokkema, 1993).

The only powdery mildew structures that are exposed for a longer period of time to mycoparasitic attack are the overwintering ascomata on the bark of woody crops such as grape-vine. However, biocontrol of grape-vine mildew through mycoparasitism by *Ampelomyces* was not effective (Falk *et al.*, 1995a). In conclusion, from an ecological perspective, there are a number of biotic and abiotic factors that do not seem favourable for the activity of *Ampelomyces* against powdery mildew fungi, yet the widespread, natural occurrence of these fungi would suggest otherwise.

Natural Occurrence and Host Range of *Ampelomyces*

Ampelomyces spp. have been reported from over 65 species (eight genera) of the Erysiphaceae from across the world (e.g., Hino & Kato, 1929; Emmons, 1930; Blumer, 1933; Clare, 1964; Belsare *et al.*, 1980; Hanlin & Tortolero, 1984; Hijwegen & Buchenauer, 1984; Sz. Nagy & Vajna, 1990; Puzanova, 1991; Tsay & Tung, 1991; Paulech *et al.*, 1993; Shin, 1994; Falk *et al.*, 1995a; Kiss 1997b; Rankovic, 1997; Kiss 1998). A 4-year study of both the natural incidence of *Ampelomyces* and the intensity of mycoparasitism in the field in a total of 27 species (nine genera) of powdery mildew fungi infecting 41 host plant genera showed that, in seven out of the nine powdery mildew genera studied, pycnidia of *Ampelomyces* were present in approximately 15–42% of samples (Kiss, 1998). However, they parasitized only approximately 1–15% of the total of the host fungal mycelia in these seven genera of the Erysiphaceae. In contrast, the natural mycoparasitism of the host mycelia was extremely high (~65%) in another powdery mildew genus, *Arthrocladiella*, represented by a single species, *A. mougeotii*, infecting *Lycium* plants (Kiss, 1998). Other field surveys also noted the widespread occurrence of *Ampelomyces* in mycelia of *A. mougeotii* on *Lycium* (e.g., De Bary, 1870; Rankovic, 1997). Surprisingly, *Ampelomyces* was rarely found in *Blumeria graminis* (DC.) Speer (Kiss, 1997b), the only powdery mildew fungus that infects monocotyledonous plants, although *in vitro* inoculation of *B. graminis* with *Ampelomyces* was as successful as it was with other species of the Erysiphaceae infecting dicotyledonous plants (Sztejnberg *et al.*, 1989; Kiss & Vajna, 1995; Mairovich *et al.*, 1996). *In vitro*, the germination of *Ampelomyces* conidia was equally stimulated by the presence of conidia of *A. mougeotii* and *B. graminis* (Gu & Ko, 1997), although, in the field, these two species were the most and the less parasitized fungal hosts among the 27 powdery mildew species examined (Kiss, 1998). So, no correlation was found between the field data and the *in vitro* mycoparasitic activity of *Ampelomyces*.

A number of other putative hosts of *Ampelomyces* reported in the older literature should also be mentioned here. Naumoff (1913) described and illustrated an *Ampelomyces* species producing internal pycnidia in the mycelia of *Bremia lactucae* Regel, the causal agent of lettuce downy mildew. However, Blumer (1933) failed to infect lettuce downy mildew with *Ampelomyces* when tried to repeat these findings. Other pycnidial fungi, identified as *Ampelomyces quisqualis*, were reported as mycoparasites of a number of species of the Mucorales (Linnemann, 1968). Similarly, a pycnidial fungus found in the hyphae of *Fusarium lateritium* Nees was described as an *Ampelomyces* species (Teterevnicova-Babajan & Nelen, 1975). These records have not been confirmed since and should be regarded as doubtful. They may represent observations made on *Phoma*-like fast growing pycnidial fungi referred to earlier in this review.

Biocontrol Potential

Yarwood (1932) was first to show the potential of *Ampelomyces* as a biocontrol agent to control clover mildew (*Erysiphe polygoni* DC) in a basic experiment, which reproduced the results of a natural epidemic recorded in the previous year. However, as with many biocontrol agents, little other work was done as fungicides were effective and were the accepted method for mildew control in most crops world-wide. It was not until the 1970s that interest increased in biocontrol, as chemical controls began to break down through the emergence of fungicide resistance. It is now becoming increasingly clear that chemical control is not always a sustainable approach, and there is world-wide concern about possible environmental drawbacks of fungicides (Fokkema, 1993). In addition, the biological control of insects, which has now become common practice in glasshouse crops, may be adversely affected by fungicide use (Dik *et al.*, 1998). As a consequence, more recent research has focused on the development of other, non-chemical, means of managing powdery mildew which can be used either in organic production or in an integrated pest management programme.

TABLE 1. A list of powdery mildew fungi and their host plants on which the biocontrol potential of different *Ampelomyces* isolates has been assessed

Crop	Powdery mildew fungus	References
Apple	<i>Podosphaera leucotricha</i>	Odintsova, 1975; Sztejnberg, 1979; Bosshard <i>et al.</i> , 1987
Black currant	<i>Sphaerotheca mors-uae</i>	Davydova <i>et al.</i> , 1985
Beetroot	<i>Erysiphe communis</i>	Puzanova, 1984
Carrot	<i>E. umbelliferarum</i> , <i>Oidium</i> sp.	Sztejnberg, 1979; Sztejnberg & Mazar, 1985; Sztejnberg <i>et al.</i> , 1989
Cucurbits	<i>Sphaerotheca</i> spp.	Jarvis & Slingsby, 1977; Philipp & Crüger, 1979; Sztejnberg, 1979; Hartmann <i>et al.</i> , 1984; Philipp <i>et al.</i> , 1982, 1984, 1990; Puzanova, 1984; Sztejnberg & Mazar, 1985; Philipp & Hellstern, 1986; Diop-Bruckler & Molot, 1987; Hijwegen, 1988, 1989; Sztejnberg <i>et al.</i> , 1989; Sundheim, 1982; Sundheim & Amundsen, 1982; Epton & Hamed El Nil, 1993; Shin & Kyeung, 1994; McGrath & Shishkoff, 1999; Dik <i>et al.</i> , 1998; Elad <i>et al.</i> , 1998; Lee, 1999
Grape-vine	<i>Uncinula necator</i>	Puzanova, 1991; Hofstein & Fridlender, 1994; Yakushkina <i>et al.</i> , 1994; Whitesides, 1994; Falk <i>et al.</i> , 1995a,b; Daoust & Hofstein, 1996; Monchiero <i>et al.</i> , 1996
Mango	<i>O. mangiferae</i>	Sztejnberg <i>et al.</i> , 1989
Mulberry	<i>Phyllactinia suffulta</i>	Sztejnberg, 1979
Peach	<i>S. pannosa</i> var. <i>persicae</i>	Marboutie <i>et al.</i> , 1993, 1995
Pepper	<i>Leveillula taurica</i>	Sztejnberg, 1979; Diop-Bruckler & Molot, 1987
Red clover	<i>E. polygoni</i> , <i>E. communis</i>	Yarwood, 1932; Puzanova, 1984
Rose	<i>S. pannosa</i> var. <i>rosae</i>	Puzanova, 1984; Whitesides, 1994; Pasini <i>et al.</i> , 1997; Verhaar <i>et al.</i> , 1999
Strawberry	<i>S. macularis</i> f. sp. <i>fragariae</i>	Nelson & Gubler, 1995
Sugar-beet	<i>E. betae</i>	Sztejnberg, 1979
Zinnia sp.	<i>Oidium</i> sp.	Sztejnberg, 1979

The first significant trial using *Ampelomyces* was reported by Jarvis and Slingsby (1977) who used conidial suspensions of the mycoparasite to control cucumber powdery mildew in greenhouse trials successfully. The control was enhanced when interspersed with water sprays. Since this work, there have been many other positive examples where *Ampelomyces* has been used to control a range of powdery mildews on several crops (Table 1), and these have paved the way for commercialisation (Hofstein & Fridlender, 1994; Daoust & Hofstein, 1996; Hofstein *et al.*, 1996). This was also supported by the lack of any acute oral toxicity of *Ampelomyces* cultures in rats. A standard eye irritation test with *Ampelomyces* in rabbits was also negative, suggesting that a fungal biocontrol product based on these mycoparasites can be used without any human health hazards in the plant protection practice (Sundheim & Tronsmo, 1988).

The most successful biocontrol experiments using *Ampelomyces* were carried out in greenhouses where the RH was kept high (e.g., Jarvis & Slingsby, 1977; Sundheim, 1982; Philipp *et al.*, 1990), or in the field where free water was frequently available on the treated leaves (e.g., Szejnberg *et al.*, 1989; Falk *et al.*, 1995b). However, the efficacy of *Ampelomyces* always decreased rapidly when the RH was below 90–95% (e.g., Philipp *et al.*, 1984; Philipp & Hellstern, 1986; Verhaar *et al.*, 1999). Clearly, the high RH requirement of *Ampelomyces* represents one of the major obstacles in its use as a reliable biocontrol agent.

Some attempts were made to enhance the efficacy of *Ampelomyces* at lower RHs. Epton & Hamed El Nil (1993) selected isolates which were able to germinate at a higher vapour deficit than the wild types. A number of additives, such as an emulsion of 1% paraffin oil (Philipp & Hellstern, 1986) or a 0.3% mineral oil surfactant (Hofstein *et al.*, 1996), were also reported to increase the biocontrol performance of *Ampelomyces* at lower RHs. However, some of these additives can control powdery mildews directly (Verhaar *et al.*, 1996; Pasini *et al.*, 1997). Shishkoff and McGrath (2002) found that Add-Q, a spray adjuvant recommended to be used together with AQ10 Biofungicide, was as effective alone against cucumber powdery mildew as combined with AQ10. Therefore, the effect of the additives and that of *Ampelomyces* should be clearly distinguished when assessing the efficacy of a biocontrol experiment. Recently, some trials have been reported where the treatments with *Ampelomyces* were ineffective, and suggested that only very limited control of powdery mildew could be expected in the plant protection practice (Dik *et al.*, 1998; McGrath & Shishkoff, 1999; Verhaar *et al.*, 1999; Shishkoff & McGrath, 2002).

In most experiments, *Ampelomyces* was sprayed onto infected plants as a conidial suspension and the applications were repeated several times during the season to ensure a high level of control (e.g., Szejnberg *et al.*, 1989; Philipp *et al.*, 1990). In order to reduce the number of applications in a vineyard, Falk *et al.* (1995b) devised a method to culture *Ampelomyces* on cotton wicks. The inoculated wicks were suspended in grape-vine trellises to allow natural dispersal of the mycoparasite in rain. This method reduced the number of applications required, but depended on the weather conditions in the vineyard. As the artificially released mycoparasites, together with the naturally occurring ones, destroyed a part of ascomata of grape-vine powdery mildew in autumn, which ensure the perpetuation of the disease in New York vineyards (Gadoury & Pearson, 1988), they contributed to a significant delay in the powdery mildew epidemic in the next season (Falk *et al.*, 1995a,b). Thus, an additional benefit of this biocontrol experiment was the reduction of the overwintering inocula of grape-vine powdery mildew.

Mass Production and Commercialization

The scale-up of *Ampelomyces* inoculum for biocontrol purposes was one of the crucial steps towards its commercialization and practical use in plant protection. Szejnberg *et al.* (1990) developed and patented (European Patent Office, publ. no. 0353662/1988) a simple and inexpensive method for production of large amounts of *Ampelomyces* conidia in fermentors. Based on this method, different formulations of *Ampelomyces* were tested in various crops,

especially in grape-vine. The early attempts to introduce a fungal biocontrol product into the grape-vine industry failed because of its poor formulation and low efficacy (Hofstein *et al.*, 1996). Finally, an improved product, AQ10™ Biofungicide, produced by Ecogen, Inc. (USA) obtained EPA registration in 1995 for use in the control of grape-vine powdery mildew. It is available as a conidial preparation dry-formulated in a stable water-dispersible granule (Daoust & Hofstein, 1996). Demonstration trials showed that spraying with AQ10 was as effective as conventional fungicides, providing that the treatment (at least two sprays) was applied when leaf incidence of mildew was still low (<10% leaf infection). The early problems related to reduced efficacy at lower RHs could be overcome by spraying in the early morning or late afternoon, and by incorporation of a mineral oil-based surfactant as a wetting agent (Daoust & Hofstein, 1996). AQ10 was subsequently registered for use on a number of other fruits and vegetables in conjunction with the wetting agent, Add-Q (Paulitz & Bélanger, 2001; Bélanger & Labbé, 2002). Currently, it is recommended as a part of an integrated programme which includes sulphur and sterol biosynthesis inhibitor fungicides. It is clear that there is still a need for more development work to produce an *Ampelomyces* product with reliability approaching that of conventional fungicides.

Mass production methods of some isolates were also developed in the former Soviet Union (Puzanova, 1984; Davydova *et al.*, 1985) where they were subsequently commercialized (Anonymous, 1989), one with the name Ampelomycin (Yakushkina *et al.*, 1994), but there is little information available about their use, efficacy and further development.

Compatibility with Fungicides

Ampelomyces has been shown to be compatible with a large number of fungicides used in the control of powdery mildews, such as triforine, quinomethionate (Sundheim, 1982; Sundheim & Amundsen, 1982), triadimefon (Philipp *et al.*, 1982; Philipp & Kirchhoff, 1983; Shishkoff & McGrath, 2002), myclobutanil (Shishkoff & McGrath, 2002) and other fungicides (Sztejnberg *et al.*, 1989; Elad *et al.*, 1996). In addition, *Ampelomyces* also tolerates many acaricides and insecticides (Philipp *et al.*, 1984), which makes it possible for their use in integrated pest management programmes.

During commercialisation, the fungal biocontrol product AQ10 was screened against a range of fungicides, insecticides and common surfactants used in grape-vine production. It was not affected by certain copper-based fungicides used against downy mildew, which allows it to be used in a single tank-mixed treatment for both diseases (Hofstein *et al.*, 1996). The fact that it is not affected by sterol biosynthesis inhibitors offers an opportunity to use the fungal product as part of an integrated pest management programme using a reduced spray regime of conventional fungicides (Hofstein & Fridlender, 1994). A benomyl-tolerant strain of AQ10 has been induced, with no detrimental effect on biocontrol activity (Feldman *et al.*, 1993), which widens the applications where this anti-powdery mildew product might be used. Interestingly, AQ10 has also been shown not to interfere with the activity of another fungal biocontrol product, Trichodex, when co-inoculated onto cucumber for dual control of powdery mildew and grey mould (Elad *et al.*, 1998).

Comparison of the Biocontrol Ability of *Ampelomyces* with that of other Antagonists

Cucumber powdery mildew is one of the major problems in greenhouse production that would require a non-chemical control method. Consequently, most of the work with fungal antagonists, including the comparison of their biocontrol efficacy, has been done against this powdery mildew disease. The screening of 17 fungal antagonists, applied as spore suspensions, showed that *A. quisqualis*, *Aphanocladium album* (Preuss) Gams and *Tilletiopsis albescens* Gotzhale were the most promising candidates for biocontrol (Hijwegen, 1988). In a semi-commercial-scale glasshouse trial, the biocontrol efficacy of *Pseudozyma flocculosa* was shown to be superior to that of the alternatives *Verticillium lecanii* (Zimm) Viegas and *A. quisqualis* (Dik *et al.*, 1998). In another experiment, isolates of *Tilletiopsis* spp., applied

24 h before inoculating cucumber plants with powdery mildew, suppressed the pathogen by 90–97% compared to the controls, while an *Ampelomyces* isolate, also applied as a prophylactic treatment, was ineffective (Hartmann *et al.*, 1984).

Rose powdery mildew is another major problem in greenhouse systems. Verhaar *et al.* (1999) compared the efficacy of five fungal biocontrol agents (*A. quisqualis*, *Aph. album*, *Pseudozyma rugulosa*, *T. minor* Nyland and *V. lecanii*) against this pathogen at different RHs, and found that only one isolate of *V. lecanii* was able to control the disease efficiently at a RH < 100%. According to Paulitz and Bélanger (2001), the best antagonist that suits the practical requirements of the cucumber and rose production in greenhouses is *P. flocculosa*.

Efficacy of *Ampelomyces* as a Biocontrol Agent

Reported data on the efficacy of *Ampelomyces* mycoparasites are contradictory. In some experiments, good control of powdery mildew infections of various crops was achieved using conidial suspensions of different isolates (Table 1) or with AQ10 (Hofstein & Fridlender, 1994; Pasini *et al.*, 1997; Romero *et al.*, 2003). Other trials showed that the biocontrol was ineffective, with or without formulation, although parasitism of powdery mildew colonies on the treated crops did occur (Diop-Bruckler & Molot, 1987; Yakushkina *et al.*, 1994; Dik *et al.*, 1998; McGrath & Shishkoff, 1999; Verhaar *et al.*, 1999; Shishkoff & McGrath, 2002). When compared with other antagonists in biocontrol trials, *Ampelomyces* isolates were less efficient than *P. flocculosa*, *P. rugulosa*, *V. lecanii*, *Aph. album* and *T. minor* (Dik *et al.*, 1998; Verhaar *et al.*, 1999). The failure of biocontrol by *Ampelomyces* in these experiments might have been due to the need for high humidity, but alternatively might have reflected the slow development of these mycoparasites in powdery mildew colonies.

CONCLUSIONS

Historically, *Ampelomyces* was one of the first mycoparasites to be studied in detail (De Bary, 1870), and was also the first fungus used as a biocontrol agent of plant parasitic fungi (Yarwood, 1932). The effect of *Ampelomyces* in the control of powdery mildew infections is slow but it is still important because it suppresses the sporulation rate of its fungal hosts, and the infected plants regain vigour after *Ampelomyces* has killed the pathogens. This mycoparasite has now become one of the most advanced in terms of commercial development of a fungal biocontrol agent for plant pathology, although there are still problems with the efficacy of the *Ampelomyces* product, AQ10 Biofungicide, especially at lower RHs. Most of the alternative fungal antagonists of powdery mildews, such as *Pseudozyma* spp., *Verticillium lecanii*, and *Tilletiopsis* spp., act more quickly through the production of antifungal compounds. However, the use of these antagonists might be hindered by the potential side-effects of their still little known metabolites (Kiss, 2003), and *Ampelomyces* might still be an appropriate candidate for future commercial development. In addition, the relationship between host plants, powdery mildews and *Ampelomyces* could be further studied from an ecological point of view to understand the role of fungal antagonists in the natural population dynamics of plant parasites.

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